

Muhammad Balawal Safdar

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github.com/muhammadbalawal

Education

Concordia University, Bachelor of Computer Science (BCS)

Sept. 2026 – Expected 2029

John Abbott College, DEC Computer Science

August 2023 – May 2026

- Dean's List recipient

Experience

Software Engineer Intern, Backboard.io

May 2026 – Present

- Built and benchmarked **Backboard's R-CLI** coding agent, which ranked **#1 on Terminal-Bench 2.1** (84.3%, solving 75/89 tasks) ahead of Codex CLI and Claude Code.
- Built **web analytics** pipelines integrating **PostgreSQL** and **Customer.io** to track and segment product usage.
- Supported hackathons including **Hack the North** through volunteer engineering with **Major League Hacking (MLH)**.

Mobile Applications Developer Intern, Mogile Technologies (ChargeHub)

January 2026 – May 2026

- Shipped Android features in **Kotlin** and **Jetpack Compose** for **ChargeHub**, an EV-charging app with **1M+ users**, including station ratings, membership features, and map functionality in an Agile team.
- Rebuilt the charging-session timer to update locally every second while syncing with server data, handling multi-session edge cases, cross-device continuity, and shared **ViewModel** state management.
- Improved engineering productivity by stabilizing flaky **Azure Pipelines** tests, authoring custom **GitHub Actions** workflows, and refactoring legacy Java into idiomatic Kotlin through peer-reviewed pull requests.

CS Tutoring, John Abbott College

March 2025 – December 2025

- Tutored 5 students in **C# programming**, focusing on OOP and data structures.
- Guided debugging, algorithm implementation, test-case design, and exam preparation.

Projects

PTAgent: Agentic AI Network Automation Tool - CUSEC 2026 Winner

[GitHub](#)

- Designed architecture using **IPC**, **Socket.IO**, and a **FastAPI** backend where an AI agent analyzes network state and executes configuration commands autonomously.
- Built an agentic AI system that automates network topology creation and configuration in **Cisco Packet Tracer**, converting a manual, multi-step workflow into automated tooling using **Python** and **JavaScript**.
- Implemented **MLOps-style deployment** patterns for the AI agent with real-time data analysis and automated workflow orchestration.

Sentinel: Traffic Accident Detection System - JACHacks 2025 Winner

[GitHub](#)

- Developed full-stack architecture integrating **MongoDB** (schema design and geospatial queries), **RESTful APIs**, **Leaflet.js** geographic clustering, **Gemini AI** incident reporting, and **Twilio** voice automation.
- Built a real-time accident detection system using **TensorFlow COCO-SSD** and a **Next.js** frontend, monitoring live Quebec 511 traffic-camera feeds with bounding-box collision detection.

Remoto: Voice-Controlled Remote Computer System - McGill 2025 Winner

[GitHub](#)

- Built a secure voice-controlled remote access system enabling real-time control of a computer from anywhere.
- Developed a custom **AI agent** that analyzes screen state using **OpenCV** and **Tesseract OCR** to generate keyboard and mouse actions via **pyautogui**.
- Implemented a **React** frontend with speech-to-text, **Supabase** (PostgreSQL schema design and authentication), and low-latency live streaming using **FFmpeg**, **MediaMTX**, and **Cloudflare**.

Skills

Languages: Python, TypeScript, JavaScript, Kotlin, C#, Java, SQL/NoSQL, PL/SQL, Bash

Platforms: Linux, Git, CI/CD, Docker, Kubernetes, AWS/Azure, JIRA, Confluence, Agile, Postman, AutoCAD

Frameworks/Libraries: Next.js, Node.js, Express.js, FastAPI, Flask, ASP.NET Core, React Native, TensorFlow

ML/AI: TensorFlow, computer vision, LLM APIs, Embeddings

Spoken Languages: English (fluent), French (professional working proficiency)

Awards & Achievements

- First Place - JACHacks 2025
- First Place - MariHacks 2026
- Second Place - McGill CodeJam 2025
- Second Place - MariHacks 2025
- Best Agentic AI - CUSEC 2026
- Best AI Project - CUHacking 2026
- Best Design - MariHacks 2026
- 11labs - JACHacks 2026
- 11labs - MariHacks 2026
- Honorable Mention - DawsHacks 2025